

- `print_operand`
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- Print functions:
 - Function to *assemble* a single instruction *assemble_insn*
 - *md_assemble* is the first function called to assemble instruction

In the *gas* directory itself the *configure.tgt* and *Makefile.am* files need to be modified to refer to the new files, and to add support for the new target.

ld

In *scripttempl/<format><target>.sc* define the default linker script file for the architecture.

In *emulparams/<target><format>.em* define the default emulation script file. This contains any functions necessary to customise the behaviour of the linker.

In *emulparams/<format><target>.sh* define any parameters that can be used to modify the default linker script file.

In the *ld* directory itself, the *configure.tgt* file needs to be updated to add your new target information and the *Makefile.am* with new target information build rules.

Build and test

Build binutils tools

Building binutils tools requires you to follow the steps shown below:

- *Configure*: Run the configure script with the target and prefix options. For example:

```
src/configure --target=cr16-elf --prefix=/local/cr16-binutils
```

Here, the *--target* option defines the target you want to build the binutils tools for, and the *--prefix* option is to specify the location of the binutils tools installation directory.

- *Build*: Run make to build the binutils tools for the above configured target. For example:

```
make all
make install
...or:
make all install
```



Note: You can use the *make -jN* option to build the tools quickly by using all processors/CPU's (that is, *N*) on your host PC. For example, if your host PC has four processors, then use *make -j4* for a quick build.

Test binutils tools

You can test the above built tools using the binutils test suites like *gas*, *binutils* and *ld*. Running the binutils test suite requires that the DejaGNU package (www.gnu.org/software/dejagnu/) is installed and the *DEJAGNU* environment variable is set. The tests can then be run with:

make check

The above command runs the binutils, gas and linker test suites.

With the following commands you can run the binutils, gas and linker test suites separately:

```
make check-binutils
make check-gas
make check-ld
```

On completion of the *binutils* test run, a summary of the results will be in the binutils directory in the *binutils.sum* file. More detailed information will also be available in the *binutils/binutils.log* file. For the gas test suite, the results are in the *gas/testsuite/gas.sum* and *gas/testsuite/gas.log* files, and for the linker test suite they are in the *ld/ld.sum* and *ld/ld.log* files.

For the most comprehensive tests in an environment where the host and target differ, DejaGNU needs some additional configuration. You can achieve this by setting the *DEJAGNU* environment variable to refer to a suitable configuration file, and defining a custom board configuration file in the directory *~/boards*. These configuration files can be used to specify a suitable simulator and how to connect it when running tests.

Documentation

Some binutils subdirectories, in turn, have the *doc* subdirectories. The documentation is written in texinfo, from which documents can be generated as PDF, PostScript, HTML or info files. The documentation is not built automatically with *make all*, nor with *make doc*. To create the documentation, change to the individual documentation directory and use *make HTML*, *make PDF*, *make ps* or *make info*, as required. The main documents of interest are:

- *bfd/doc/bfd.texinfo* is the BFD manual
- *binutils/doc/binutils.texi* is the main binutils user guide
- *ld/ld.texinfo* is the linker user guide
- *gas/doc/as.texinfo* is the assembler user guide

The exception to automatic building is with *make install*. This will build info files for any documents in the *binutils/doc* directory and install them in the *info* subdirectory of the install directory.

Note: This document is based on the author's experience till date. It will be updated in future. Suggestions for improvements are always welcome. 

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The author has years of experience porting GNU tools to different RISC/DSP processors. He is currently the maintainer of National Semiconductors' 32-bit and 16-bit processors of binutils and GDB simulator ports. You can contact him at MR.Swami.Reddy@nsc.com for further comments/suggestions.